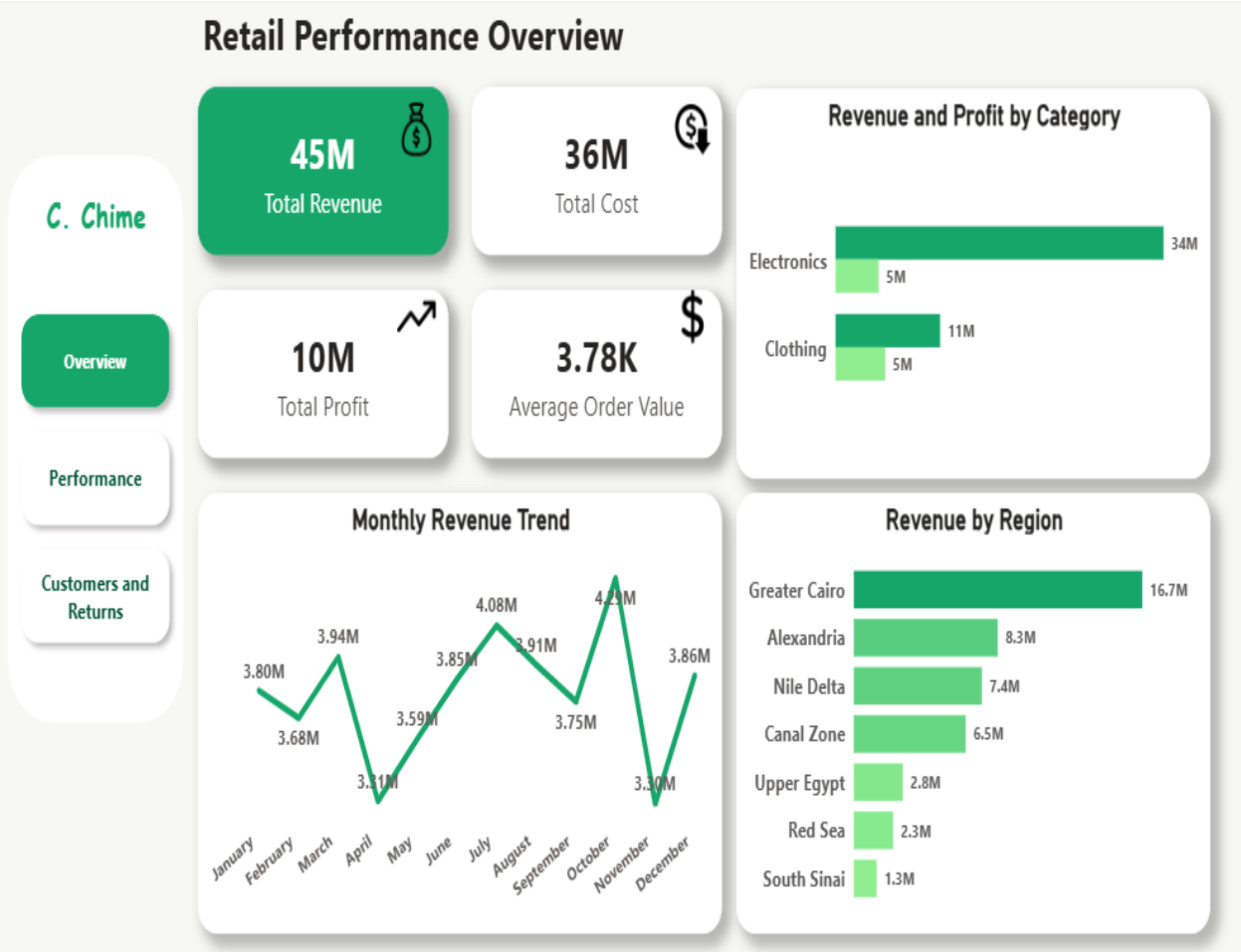
Retail Sales, Customer, and Operations Analysis

ABSTRACT


Retail Performance Overview
C. Chime
Overview
Performance
Customers and Returns
45M
Total Revenue
36M
Total Cost
10M
Total Profit
3.78K
Average Order Value
Revenue and Profit by Category
Electronics
34M
5M
Clothing
11M
5M
Monthly Revenue Trend
3.80M
3.68M
3.94M
3.31M
3.59M
3.85M
4.08M
3.91M
3.75M
4.21M
3.30M
3.86M
January
February
March
April
May
June
July
August
September
October
November
December
Revenue by Region
Greater Cairo
16.7M
Alexandria
8.3M
Nile Delta
7.4M
Canal Zone
6.5M
Upper Egypt
2.8M
Red Sea
2.3M
South Sinai
1.3M

Product, Category and Store performance

C. Chime

Overview

Performance

Customers and Returns

43K
Total Quantity Sold

21.4%
Profit Margin

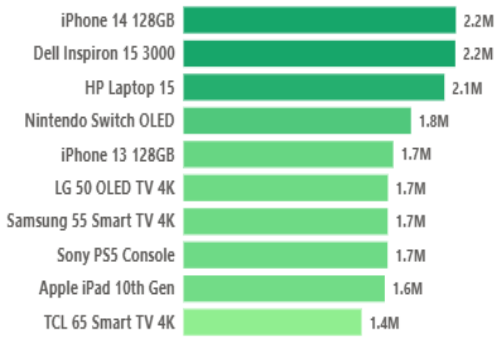
7.9%
Average discount

12K
Total Orders

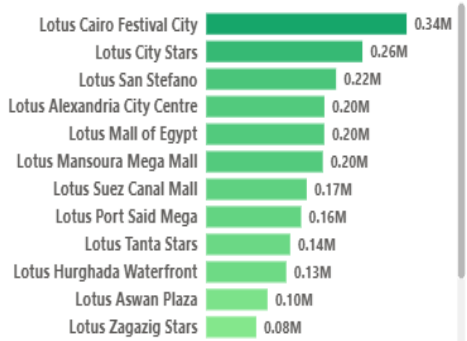
Regional Performance

Region	Total Revenue	Total Cost	Total Profit	Profit Margin
Upper Egypt	2,829,136	2,204,320	624,816	22.1%
South Sinai	1,320,701	1,048,700	272,001	20.6%
Red Sea	2,270,526	1,809,720	460,806	20.3%
Nile Delta	7,414,450	5,819,030	1,595,420	21.5%
Greater Cairo	16,706,829	13,080,420	3,626,409	21.7%
Canal Zone	6,476,891	5,106,310	1,370,581	21.2%
Alexandria	8,332,447	6,579,650	1,752,797	21.0%
Total	**45,350,979**	**35,648,150**	**9,702,829**	**21.4%**

Revenue by Product



Store Profitability



Customers and Returns Analysis

C. Chime

Overview

Performance

Customers and Returns

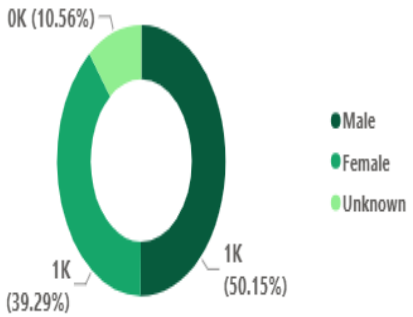
3K
Unique Customers

4.01M
Total Return Amount

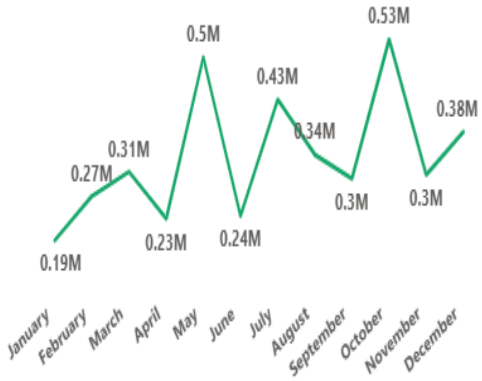
8.8%
Return Rate

15.40K
Revenue per Customer

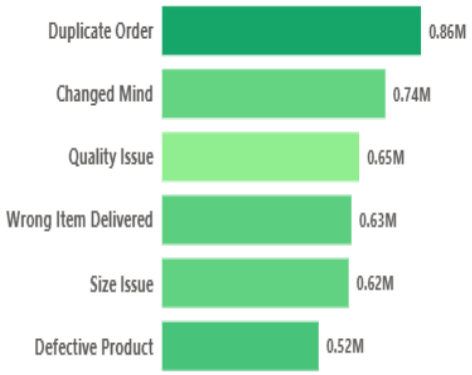
Customer Distribution



Return Amount Trend



Return Amount by Return Reason



INTRODUCTION

Background

The retail industry generates volumes of data across customers, products, stores, employees, sales transactions, and product returns. When properly analyzed, this data can provide, valuable insights into purchasing behavior, revenue performance, customer loyalty, store efficiency, and operational challenges.

This analysis uses an integrated retail dataset containing customer demographics, store information, employee details, product attributes, sales transactions, dates and returns which makes it possible to examine business performance from multiple perspectives including what products are selling, where revenue is being generated, who the most valuable customers are, how stores and employees are performing and where returns or discounts may be affecting profitability.

Problem Statement

Despite having access to detailed sales, customer, product, store, employee and returns data, retail businesses may struggle to convert this information into actionable insights. There is a need to analyze available data to uncover patterns and performance gaps across sales, customers, products, stores, employees, discounts and returns. The findings will help management understand the key drivers of revenue and profitability and make informed decisions that improve customer retention, operational efficiency, sales performance, and overall business growth

Research Questions

1. What are the key drivers of revenue and profitability and how do they vary across products, categories, stores, regions, and time?

2. How do sales volume, order activity, costs and discount influence overall revenue and profitability?

3. Who are the customers and how does customer value and composition vary across demographics?

4. What patterns exist in product returns across time and regions, and what do they imply for business performance?

5. How can insights from product, financial, customer and return performance be used to improve business decision-making and operational efficiency?

Objectives

1. To evaluate the key drivers of revenue and profitability.

2. To identify top performing products, categories and stores.

3. To evaluate sales volume, order activity, and costs.

4. Examine customer composition and value.

5. Analyze product return patterns.

DATA DESCRIPTION

Data Sources: The dataset is from an external data source form a retail sales database. The data was organized into fact and dimension tables to support detailed analysis.

Data Characteristics: The dataset contains both transactional and descriptive data. The fact tables capture business activities such as orders, product sales, and returns, while the dimension tables provide details about customers, products, stores, employees and dates. The data covers different periods and includes numerical, categorical and date fields.

METHODOLOGY

Data Cleaning

1. Duplicate Removal: In my data cleaning procedure, I checked key fields such as employee id, product id and returns id for duplicate records and removed confirmed duplicates.

2. Handling missing values: Reviewed blank fields and replaced or handled them appropriately where necessary.

3. Standardizing text: Removed unnecessary spaces and standardized text formats such as names, cities, regions, and categories.

4. Data Types: Ensured dates, IDs, numerical values, and categorical fields had the appropriate data types.

5. **Checking consistency**: Reviewed relationships between fact and dimension tables and ensured key fields were suitable for establishing relationships.

6. **Checking consistency**: Checked for incorrect or inconsistent values to ensure the dataset was reliable for analysis.

Exploratory Data Analysis

These measures were used to explore patterns and relationships across products, categories, stores, region, customers, discounts, returns and time.

a. Average Selling Price (1,050): Measures the average selling price of products.

b. Average Unit Cost (821.27): Measures the average cost incurred per product unit.

c. Revenue YoY Growth (52.3%): Measures the percentage change in revenue compared with the previous year.

d. Profit YoY Growth (1,050): Measures the percentage change in profit compared with the previous year.

e. Revenue Contribution (100%): Shows the percentage of revenue contributed by a particular product or category.

f. Profit Contribution (100%): Shows the percentage of profit contributed by a particular product or category.

g. Product Revenue Rank: Ranks products according to their revenue contribution, from highest to lowest.

h. Product Profit Rank: Ranks product according to the product they generate.

i. Store Revenue Rank: Ranks stores based on the revenue they generate.

j. Store Profit Rank: Ranks stores based on their profitability.

k. Total Returns (1000 units): Counts the number of return transactions.

l. Returned Orders (1000): Counts the number of unique orders associated with returns.

m. Return Amount % of Revenue (0.09%): Shows the value of returns relative to total revenue.

Key Performance Indicators

a. Total Revenue: Measures the total value generated from product sales

b. Total Cost: Measures the total cost incurred in providing the products sold.

c. Total Profit: Calculates the difference between total revenue and total cost, showing the amount earned after costs.

d. Profit Margin: Measures profit as a percentage of revenue, indicating how efficiently sales are converted into profit.

e. Total Quantity Sold: Shows the total number of unit products sold.

f. Total Orders: Counts the number of unique customer orders.

g. Unique Customers: Measures the number of distinct customers who made purchases.

h. Average Order Value: Shows the average revenue generated per order.

i. Total Return Amount: Measures the total monetary value of returned products.

j. Revenue per Customer: Measures the average revenue generated by each customer.

k. Return Rate: Measures the proportion of orders that resulted in returns.

l. Average Discount: Average percentage discount applied to sales.

ANALYSIS AND FINDINGS

Analysis Questions

1. What are the key drivers of revenue and profitability and how do they vary across products, categories, stores, regions, and time?

The Revenue and Profit by Category chart shows the contribution of the product category to revenue and profit. Electronics is the strongest revenue-generating category, recording approximately 34million in revenue, compared with approximately 11million for clothing. Both categories contribute approximately 5 million in profit.

The Revenue by product chart identifies the individual products generating the highest revenue. The leading products include major technology products such as iPhones, HP laptops, Nintendo Switch, LG television, Samsung Products, Sony PlayStation and Apple iPads.

The Revenue by Region chart shows clear differences in revenue performance across geographical regions. Greater Cairo is the strongest performing region generating approximately 16.7million, followed by Alexandria at approximately 8.3million and Nile Delta at approximately 7.4million. The remaining regions contribute lower levels of revenue.

The Regional Table on the performance page provides a deeper comparison by showing Total Revenue, Total Cost, Total Profit and Profit Margin.

Greater Cairo records approximately: Revenue (16.7M), Cost (13.1M), Profit (3.63M), Profit Margin (21.7%). However Greater Cairo does not have the highest profit margin. **Upper Egypt records the highest displayed margin at 22.1%.** This is an important finding because it shows that the region generating the most revenue is not necessarily the region operating at the highest profit efficiency.

The Store Profitability chart shows differences in profit across individual stores. Lotus Cairo Festival City Store has the highest displayed profit of 0.3M (#335,435). Followed by Lotus City Stars at approximately 0.2M (#261,445). Other stores

generate progressively lower levels of profit. This indicates that store location and individual store performance contribute to overall profitability.

The Monthly Revenue Trend shows how revenue changes throughout the year. Revenue fluctuates rather than remaining constant. The highest value occurs in October at approximately 4.2M, while some of the lowest points occur around April and November at approximately 3.3M. This demonstrates that sales performance changes over time.

2. How do sales volume, order activity, costs and discount influence overall revenue and profitability?

This question is mainly answered by the performance page. This analysis records approximately;

a. 43K units sold

b. 12K orders

c. 45.35M revenue

d. 36M cost

e. 9.70M profit

f. 21.4% profit margin

g. 7.9% average discount

These indicators provide an overall view on how sales activity translates into financial performance. The business generated substantial sales volume and revenue, but a significant proportion of revenue was absorbed by cost. The resulting profit margin of 21.4% indicates that the business remains profitable overall

3. Who are the customers and how does customer value and composition vary across demographics?

The customer distribution chart shows the composition of customers by gender: Male (1477), Female (1157), Unknown (311). This indicates that male customers represent the largest identified customer group, while female customers also make up a substantial share of the customer base. The presence of the unknown category is also important because it shows that gender information is not available for all customers.

4. What patterns exist in product returns across time and regions, and what do they imply for business performance?

The business records approximately 4.01M in total return amount and an overall 8.8% return rate.

The Return Amount by Return Reason chart identifies the main reasons contributing to the financial value of returns:

Duplicate order – 0.8M

Changed mind – 0.74M

Quality issues – 0.65M

Wrong item delivered – 0.63M

Size Issue – 0.62M

Defective product – 0.52M

The largest return amount comes from Duplicate orders followed by Changed mind.

The Return Amount Trend shows that return fluctuates throughout the year. October records the highest displayed return amount at approximately 0.53M, while other notable peaks occur around May and July.

5. How can insights from product, financial, customer and return performance be used to improve business decision-making and operational efficiency?

The dashboard shows that Electronics and Greater Cairo are major revenue contributors, while differences in regional profit margins highlight the importance of profitability efficiency. The customer analysis provides information about customer composition, while the returns analysis identifies both the financial impact of returns and the major reason behind them.

Key Performance Indicators

For my analysis, I had the following Key Performance Indicators

a. Total Revenue (45.35M): This is the total value of sales generated during the analysis period.

b. Total Cost (36M): This is the total cost associated with the recorded sales.

c. Total Profit (9.70M): This is the remaining revenue after deducting total cost.

d. Average Order Value (3.78k): This is the average revenue generated per order.

e. Total Quantity Sold (43K): Total of product units sold

f. Profit Margin (21.4%): Percentage of revenue retained as profit after costs

g. Average Discount (7.9%): Average percentage discount applied to sales.

h. Total Orders (12k): Total number of orders recorded

i. Total Return Amount (4.01million): This represents the total monetary value associated with returned transactions.

j. Revenue per Customer (15,400): An average of approximately 15.40K per customer provides an indication of customer value based on revenue.

k. Return Rate (8.8%): This means that returns represent a measurable share of the business's sales activity.

l. Unique Customers (3K): This represents the number of distinct customers who made purchases during the analysis period.

RESULT

Summary of Findings

The three dashboard pages reveal that the business is profitable but has considerable variation in performance across products, categories, regions, stores, customers and time.

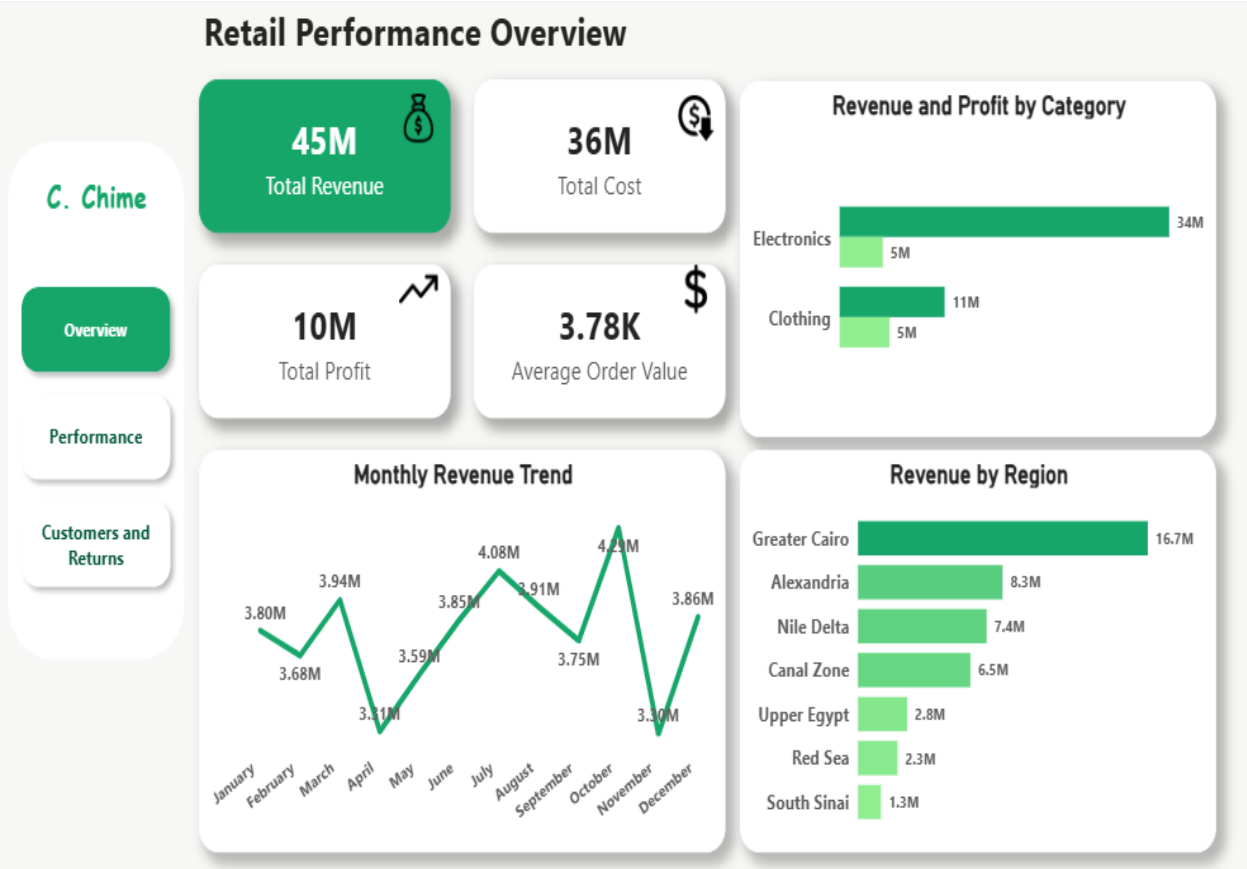
The major findings are:

- Total revenue is approximately 43.35M
- Total cost is approximately 36M
- Total profit is approximately 9.70M
- Overall profit margin
- Approximately 43K units were sold through 12K orders.
- Average Order Value is approximately 3.78K.
- Electronics is the leading revenue-generating category at approximately 34M.
- Greater Cairo is the strongest revenue-generating region at approximately 16.7M.
- Cairo has the highest displayed store profit at approximately 0.60M.
- Upper Egypt has the highest displayed regional profit margin at 22.1%.
- Revenue fluctuates across the year, with October recording the highest visible monthly revenue.
- There are approximately 3K unique customers.
- Male customers represent 50.15%, females 39.29%, and unknown gender 10.56%.
- Revenue per customer is approximately 15.40K.
- Total return amount is approximately 4.01M.
- Overall return rate is 8.8%.
- Return amounts fluctuate throughout the year, with October recording the highest visible return amount of approximately 0.53M.
- Duplicate Orders are the largest return reason by return amount at approximately 0.86M.

- Changed Mind is the second-largest return reason at approximately 0.74M.
- Quality Issues, Wrong Item Delivered, Size Issues and Defective Products also contribute substantial return amounts.
- Average discount is 7.9%.

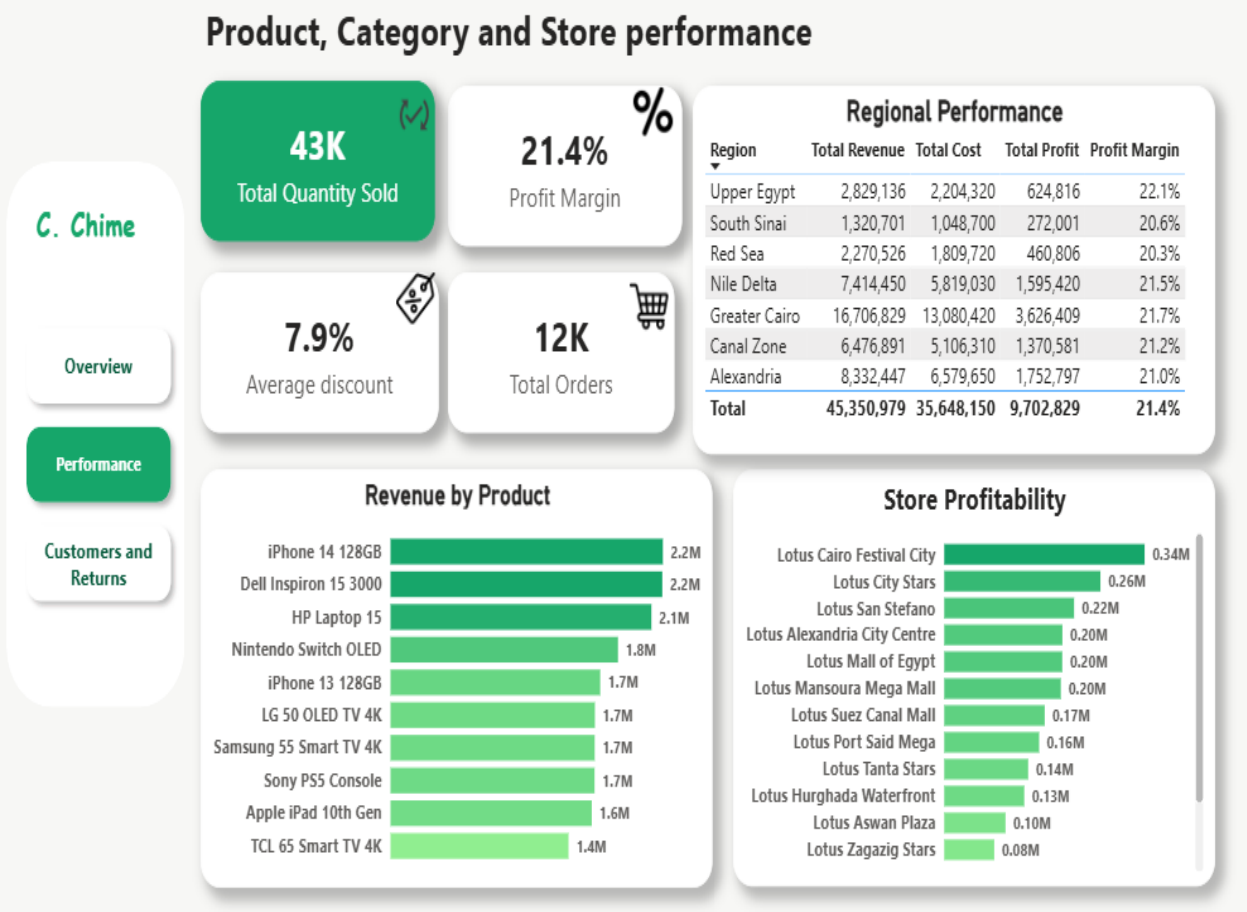
DASHBOARD

Page 1: Retail Performance Overview



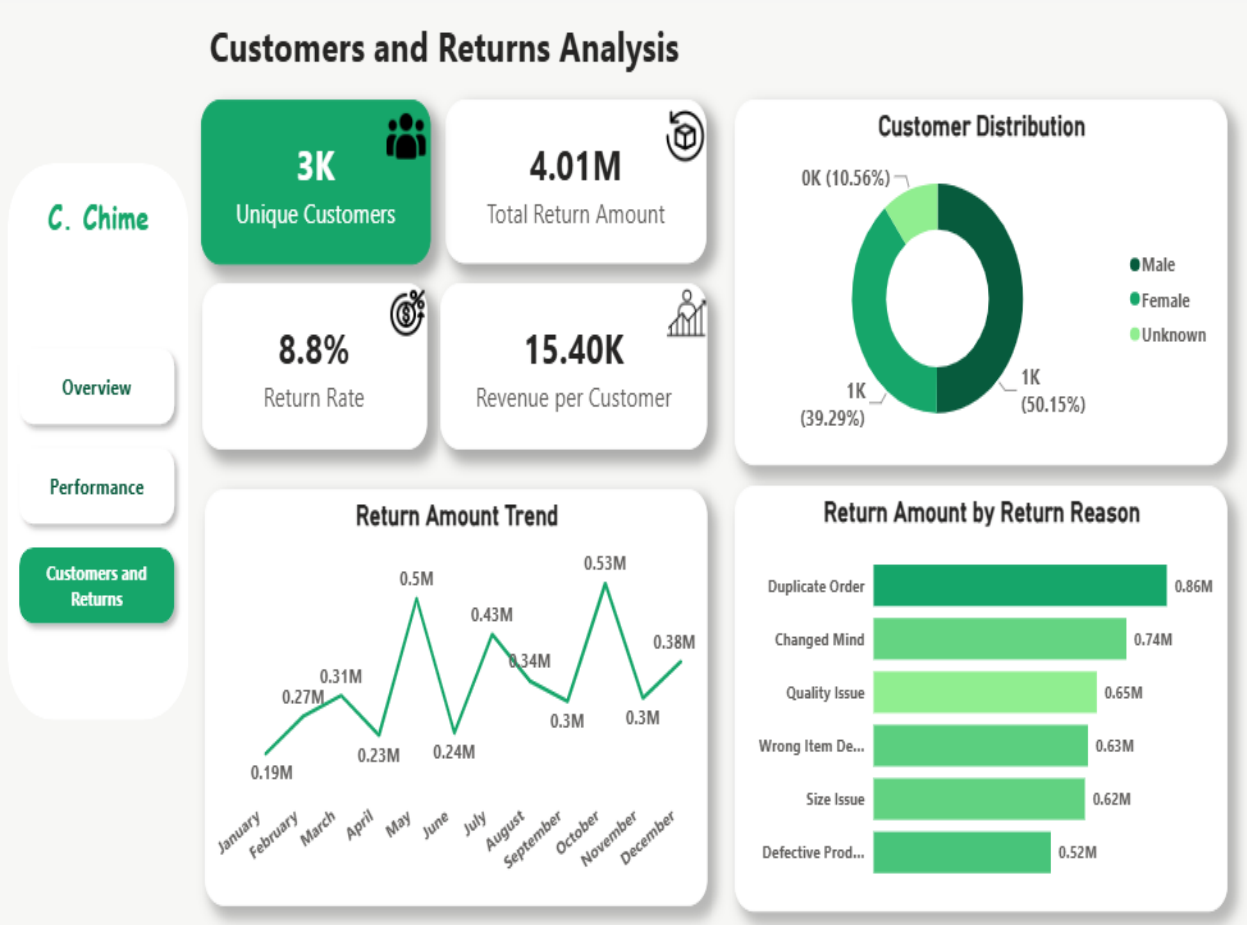
The Retail Performance Overview page provides a high-level view of the business's overall performance. It presents total revenue, total cost, total profit and average order value, while the charts show revenue and profit by category, monthly revenue trends and revenue across regions. The page helps identify major revenue drivers, track changes in sales over time and compare performance across geographical regions.

Page 2: Product, Category and Store Performance



The Product, Category and Store Performance page provides a more detailed assessment of sales and profitability. It presents total quantity sold, profit margin, average discount and total orders, while the visuals compare regional revenue, cost, profit and profit margin, identify the highest-revenue products and show profitability across stores. The page helps determine which products, regions and stores are performing best and how efficiently sales are being converted into profit.

Page 3: Customers and Returns Analysis



The Customers and Returns Analysis page focuses on customer composition, customer value and product returns. It presents unique customers, total return amount, return rate and revenue per customer, while the charts show customer distribution by gender, return amounts over time and return amounts by return reason. This page helps the business understand its customer base, assess customer value, identify periods of high returns and determine the main reasons responsible for the financial impact of returns.

DISCUSSION

Interpretation of Results

RQ1 — Revenue and Profitability Drivers

The results show that product category, individual products, geographical region, store location and time all contribute to differences in performance. Electronics and Greater Cairo are major revenue drivers, while Upper Egypt achieves the highest displayed regional profit margin. This shows that management should consider both revenue generation and profitability efficiency when assessing performance.

RQ2 — Sales, Costs, Discounts and Profitability

The business generated 45.35M in revenue and 9.70M in profit from approximately 43K units and 12K orders. With an average discount of 7.9% and profit margin of 21.4%, the results show that the business remains profitable while using discounts as part of its sales strategy. However, costs and discount levels need to be carefully monitored to maintain healthy margins.

RQ3 — Customer Value and Composition

The customer base consists of approximately 3K unique customers, with males representing the largest identified customer group. Revenue per customer of 15.40K provides an indication of customer value. These findings can support more targeted customer engagement and retention strategies.

RQ4 — Return Patterns and Causes

Returns represent approximately 4.01M in value and an 8.8% return rate. Return values fluctuate over time, with October recording the highest displayed amount. Duplicate Orders are the largest contributor to return value, followed by Changed Mind, Quality Issues and Wrong Item Delivered. This indicates that some returns may be reduced through improvements in the ordering and fulfilment processes, while others may require better product information, quality control or customer communication.

RQ5 — Business Decision-Making

The combined analysis demonstrates that management can make better decisions by considering multiple performance indicators together. Revenue, profitability, customer value and return reasons provide complementary information that can guide decisions on inventory, pricing, marketing, customer retention, quality control and operational efficiency.

Implication of findings

The findings show that the business is performing profitably, but performance is not evenly distributed across products, regions, stores and time. The strong contribution of Electronics and Greater Cairo suggests that these areas are important to overall business performance and should be carefully managed. However, the higher profit margin recorded in Upper Egypt shows that high revenue does not always mean greater profitability efficiency.

The analysis also shows that returns are a significant area of concern, with a total return amount of 4.01M and a return rate of 8.8%. The high return value associated with duplicate orders, changed minds, quality issues and wrong items delivered suggests that both operational and customer-related factors contribute to returns. Reducing avoidable returns could therefore protect revenue and improve operational efficiency.

Finally, the customer and discount findings indicate opportunities for more targeted decision-making. Customer composition and revenue per customer can support better marketing and retention strategies, while the 7.9% average discount should be monitored to ensure that promotional activities increase sales without putting unnecessary pressure on profit margins.

CONCLUSION

Summary

This analysis provides a comprehensive view of retail performance across financial performance, products, categories, regions, stores, customers and returns.

The business generated approximately 45.35M in revenue and 9.70M in profit, resulting in an overall profit margin of 21.4%. Electronics is the largest revenue-generating category, while Greater Cairo is the strongest region in terms of revenue and absolute profit. However, Upper Egypt records the highest displayed regional profit margin, showing that high revenue does not always translate into the highest profitability efficiency.

The customer analysis shows approximately 3K unique customers, with males representing the largest identified customer group and average revenue per customer of 15.40K.

The returns analysis records approximately 4.01M in return value and an 8.8% return rate. Importantly, the analysis identifies Duplicate Orders as the largest contributor to return value, followed by Changed Mind, Quality Issues and Wrong Item Delivered. This makes the return analysis more actionable because it identifies potential areas for operational improvement rather than simply showing the volume of returns.

Overall, the dashboard demonstrates that effective retail management requires a balance between revenue growth, profitability, customer value and return management. The findings provide a data-driven basis for improving inventory decisions, pricing, customer strategies, operational processes and overall business efficiency.

Recommendations

1. Strengthen Return Management

-Investigate the high value of returns caused by duplicate orders.

-Improve order confirmation and processing procedures to minimize duplicate orders.

-Analyze quality and defective-product returns and address recurring product issues.

-Improve picking, packing and delivery processes to reduce wrong-item deliveries.

2. Improve Profitability and Cost Control

-Evaluate products, stores and regions using both revenue and profit margin.

-Identify practices in high-margin regions and stores that can be replicated elsewhere.

-Regularly monitor operating costs to protect profit margins.

-Avoid pursuing sales growth at the expense of profitability.

3. Optimize Pricing and Discount Strategies

-Monitor the effect of the 7.9% average discount on revenue and profit.

-Use discounts strategically for products or periods where they are most likely to increase sales.

-Review products with high sales but relatively low profitability.

-Set appropriate discount limits to prevent unnecessary erosion of margins.

4. Strengthen Customer-Focused Strategies

-Use customer value and demographic information to develop targeted marketing campaigns.

-Focus retention efforts on customers who generate higher revenue.

-Improve product information and communication to reduce returns caused by changed minds and size issues.

-Maintain accurate customer information to improve the quality of future analysis.

5. Use Data-Driven Operational Planning

-Use monthly revenue and return trends to improve inventory and staffing decisions.

-Prioritize high-performing products and regions when allocating resources.

-Regularly monitor store and regional performance through the dashboard.

-Use dashboard insights as part of periodic management reviews and strategic planning.